



DHEIS – District Health & Education Intelligence System

The domain of the Project:
DATA ANALYTICS

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Period of the project
August 2026 to September 2026



DECLARATION

The project titled “**District Health & Education Intelligence System (DHEIS)**” has been mentored by **Nithyasri kannathal**, organised by **SURE Trust**, from **August 2026 to September 2026**, for the benefit of educated unemployed rural youth for gaining hands-on experience in working on industry-relevant projects that would take them closer to prospective employers.

I declare that, to the best of my knowledge, the project work mentioned below has been successfully completed by me and has helped me enhance my practical knowledge and skills in the domain of **Data Analytics, Data Engineering, Machine Learning, Business Intelligence, and Generative AI**.

The project was completed individually with guidance from the mentor. AI tools were also used as an assistance during the learning and development process, while the implementation, analysis, testing, and final project work were completed and reviewed by me.

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1. EXECUTIVE SUMMARY

A. Project Overview

DHEIS (District Health & Education Intelligence System) is an integrated data analytics and decision-support system developed to identify districts that require relatively greater attention based on health and education conditions. The project combines district-level health and education information into a single analytical framework so that users can understand overall district priority, identify the major areas of concern, compare districts, and support evidence-based resource allocation.

The motivation behind DHEIS comes from the difficulty of analysing health and education conditions when the information is available across separate datasets and contains indicators with different units, scales and directions. A district may perform well on one indicator but show considerable vulnerability on another. Therefore, looking at individual indicators independently may not provide a complete picture of district-level need. DHEIS addresses this problem by integrating the two domains and converting selected indicators into a common vulnerability-based framework.

The project follows an end-to-end analytical workflow:

Raw Data → Data Cleaning → Data Validation → Data Engineering → Analysis → Priority Index → Machine Learning Validation → Dashboard → Application → AI Intelligence → Resource Allocation

The final objective is not simply to display data, but to convert data into understandable intelligence that can support better prioritisation and planning.

B. Core Objectives and Methodology

The primary objective of DHEIS is to develop a transparent and practical system for district-level prioritisation using integrated health and education indicators.

The project uses health information from **NFHS-5** and education information from **DISE**. The datasets were collected, examined, cleaned and reconciled at district level. After the integration and final reconciliation process, **602 districts** containing the required health and education information were used for the core analysis.

The project uses a structured data engineering architecture consisting of **PostgreSQL/Supabase, Databricks, Apache Spark/PySpark and Delta Lake**. A



Bronze-Silver-Gold approach was implemented to separate raw data, cleaned/transformed data and business-ready analytical outputs.

During data validation, an important issue was identified with the available pupil-teacher ratio field. The values did not behave as expected for a meaningful pupil-teacher ratio. Instead of directly accepting the field, the underlying education variables were investigated and the ratio was recalculated using:

True Pupil-Teacher Ratio = Total Enrollment / Total Teachers

This correction was incorporated into the analytical dataset after validation.

For prioritisation, fourteen Tier 1 indicators were selected from the health and education domains. Because the indicators have different units and directions, they were transformed into a common **0–100 vulnerability scale**. Higher-is-better indicators were reversed so that a higher vulnerability value consistently represented greater concern.

The eight health indicators were combined to create a **Health Sub-Score**, while six education indicators were combined to create an **Education Sub-Score**. These two scores were then combined using:

DHEIS Priority Index = (Health Sub-Score + Education Sub-Score) / 2

Based on the resulting relative priority distribution, districts were classified into **High, Medium and Low priority tiers**. The system also identifies whether the relative vulnerability is primarily **Health-driven, Education-driven, or Both equally**.

Machine learning was then introduced as a validation layer. The original fourteen Tier 1 indicators were used as model features, while derived variables such as the Priority Index, Health Sub-Score, Education Sub-Score and Dominant Driver were excluded to prevent target leakage. Dummy Classifier, Logistic Regression, Random Forest and Gradient Boosting were evaluated using stratified validation and model tuning.

The final selected model was a **tuned Logistic Regression model**, which achieved **95% accuracy** on the held-out test set and achieved **96% precision, 96% recall and 96% F1-score for the High-priority class**.

The analytical outputs were then made available through two major user-facing platforms: **Power BI** and **Streamlit**. Power BI provides interactive dashboards for priority distribution, health and education performance, district comparisons, indicator analysis, drivers and action-oriented views. Streamlit provides a more interactive district intelligence experience, including district-versus-state benchmarking, two-district comparison and indicator-gap analysis.

A generative-AI layer was also integrated into the Streamlit application using the **Groq API** and the **openai/gpt-oss-20b** model. The AI layer generates structured



district briefs and two-district comparison briefs using the project data supplied to it. It is constrained from inventing statistics, unsupported causes, policies or causal conclusions.

C. Project Technology Stack

Component	Technology Used	Purpose
Data Sources	NFHS-5, DISE	Health and education information
Database	PostgreSQL / Supabase	Structured data storage
Data Engineering	Databricks	Data processing and transformation
Processing	Apache Spark / PySpark	Large-scale data processing
Storage Architecture	Delta Lake	Reliable analytical data storage
Programming	Python	Data analysis, ML and application development
Data Analysis	Pandas	Data preparation and exploration
Visualization	Power BI, Plotly	Interactive analytical reporting
Machine Learning	Scikit-learn	Priority-tier validation
MLOps	MLflow	Experiment and model tracking
Application	Streamlit	Interactive district intelligence
Generative AI	Groq API	AI-assisted interpretation
Version Control	Git / GitHub	Project and code management



D. Key Analytical Innovation

The central analytical innovation of DHEIS is the **DHEIS Priority Index**.

Instead of presenting fourteen different indicators independently, DHEIS converts them into a common vulnerability scale and combines them into two interpretable domain-level scores.

Health Indicators → Health Sub-Score

Education Indicators → Education Sub-Score

Health Sub-Score + Education Sub-Score → DHEIS Priority Index

This allows the system to answer a more meaningful question:

Which districts require relatively greater attention when both health and education conditions are considered together?

The system then adds a second level of interpretation by identifying the dominant domain contributing to the relative vulnerability:

- **Health-driven**
- **Education-driven**
- **Both equally**

This makes the Priority Index more useful than a simple ranking because it also provides an initial indication of where the main concern lies.

E. Key Findings and Results

The major results obtained from the completed DHEIS workflow are summarised below.

Project Measure	Result
District dimension records	707
Raw health records	706
Raw education records	603
Final integrated districts	602
Tier 1 indicators	14
Health Tier 1 indicators	8



Education Tier 1 indicators	6
Priority classification	High / Medium / Low
Machine learning test districts	121
Final ML model	Tuned Logistic Regression
Test Accuracy	95%
High-tier Precision	96%
High-tier Recall	96%
High-tier F1-score	96%
Production tier matches	577 of 602
Production tier differences	25 of 602
Registered MLflow model	district_priority_classifier

The machine learning results indicate that the selected model can reproduce the calculated priority classification with high performance on the held-out test data. However, the ML model is used as a validation component rather than as a replacement for the transparent DHEIS scoring methodology.

F. From Identification to Action

A major objective of DHEIS is to move beyond simply identifying high-priority districts.

The system follows a decision-support workflow:

Identify → Diagnose → Prioritize → Allocate → Intervene

Identify

The Priority Index identifies districts that fall into the High, Medium or Low relative-priority categories.

Diagnose

The health and education indicators are examined to understand the major areas contributing to the district's vulnerability.



Prioritize

The system identifies the dominant driver and highlights areas requiring greater attention.

Allocate

Action-oriented dashboard views support resource-allocation thinking through urgency levels and budget strategies.

Intervene

The resulting information can be used by human decision-makers to consider appropriate district-level interventions.

DHEIS therefore connects analytical findings with potential action rather than ending at data visualization.

G. AI-Assisted District Intelligence

The project also includes a generative-AI component designed to make analytical results easier to understand.

For an individual district, the AI generates a structured brief containing:

1. Overall Assessment
2. Key Concerns
3. Priority Areas
4. Recommendations

For comparing two districts, the system generates:

1. Overall Assessment
2. Key Differences
3. Priority Areas
4. Recommendations

The AI layer is designed as an **evidence-grounded interpretation assistant**. It receives structured project information and generates explanations based only on the supplied data.



The system is instructed not to:

- invent statistics,
- introduce unsupported causes,
- create unsupported government policies,
- make causal claims,
- create unsupported targets,
- or present AI-generated suggestions as guaranteed outcomes.

Therefore, the AI component supports interpretation while keeping final decisions with human users.

H. Strategic Value of DHEIS

DHEIS demonstrates how several modern data technologies can be combined into one end-to-end system:

Data Engineering + Data Analytics + Machine Learning + MLOps + Business Intelligence + Application Development + Generative AI

The project transforms fragmented source data into a structured decision-support workflow. Instead of requiring a user to inspect multiple datasets independently, DHEIS provides an integrated view of district conditions, relative priority, major drivers and possible areas for action.

The system is particularly relevant to situations where decision-makers need to compare many districts and identify where limited attention or resources may be directed first.

DHEIS is not intended to replace government officials, administrators, analysts or policy experts. Its purpose is to organise evidence, simplify comparison and provide analytical support for human decision-making.

I. Overall Project Outcome

The completed DHEIS system demonstrates an end-to-end transformation from raw data to actionable intelligence.



Raw Health & Education Data



Clean and Validate



Integrate District Information



Process through Bronze-Silver-Gold Architecture



Analyse Indicators



Calculate DHEIS Priority Index



Classify District Priority



Validate Using Machine Learning



Visualise in Power BI



Explore Through Streamlit



Generate Evidence-Grounded AI Briefs



Support Resource Allocation and Intervention Planning

The final outcome is therefore not a single dashboard or machine-learning model. DHEIS is an integrated district intelligence system that brings together data engineering, analytical scoring, machine learning, interactive visualization and AI-assisted interpretation.

The project demonstrates the complete journey from:

Data → Information → Intelligence → Priority → Action



2. INTRODUCTION

A. Background and Context

Health and education are two of the most important areas that influence the social and economic development of a district. Both areas contain multiple indicators that can be used to understand the conditions experienced by different communities. However, these indicators are often available through separate datasets and may use different measurement scales, definitions and directions.

For example, a higher immunization percentage generally represents better performance, while a higher dropout rate represents greater concern. Similarly, literacy, sanitation, electricity, water availability and road connectivity provide different perspectives on the education and health environment of a district. Analysing these indicators individually can make it difficult to determine which districts require relatively greater attention when several indicators are considered together.

DHEIS (District Health & Education Intelligence System) was developed to address this analytical challenge by bringing district-level health and education information into one integrated decision-support framework.

The project uses health indicators from **NFHS-5** and education indicators from **DISE**. These datasets were processed, validated and integrated at district level to create a common analytical base. The final reconciled dataset contains **602 districts** with the required health and education information.

The purpose of DHEIS is not simply to display these indicators. It is to transform them into understandable intelligence that can help users identify relative priority, understand major areas of concern and support resource-allocation decisions.

B. The District-Level Decision Problem

A major challenge in district-level planning is determining **where attention should be prioritised when resources and interventions are limited**.

A district may have strong performance in some areas while experiencing weaknesses in others. Looking only at a single indicator may therefore provide an incomplete picture. For instance, a district may have relatively good literacy but poor health-related outcomes. Another district may have better health conditions but face serious education infrastructure or dropout concerns. A meaningful decision-support system should be able to consider these conditions together rather than treating every indicator as an isolated problem.

The problem addressed by DHEIS can therefore be expressed as:

How can district-level health and education indicators be integrated and transformed into a transparent priority framework that identifies relatively higher-need districts and explains the major areas contributing to that priority?



DHEIS addresses this problem through data integration, indicator normalization, priority scoring, machine-learning validation and interactive analytical applications.

C. Need for Integrated Health and Education Intelligence

Traditional analytical approaches often require users to examine multiple datasets, calculate comparisons manually and interpret indicators separately. This can make district-level comparison time-consuming and difficult, especially when the number of indicators is large. DHEIS introduces an integrated approach in which health and education indicators are processed through a common analytical workflow.

The system converts selected indicators into a **0–100 vulnerability scale**. Higher-is-better indicators are reversed so that the resulting values follow a consistent interpretation: a higher vulnerability score represents greater relative concern.

The indicators are then grouped into two major domains:

Health Domain → Health Sub-Score

Education Domain → Education Sub-Score

These are combined to produce the:

DHEIS Priority Index

This provides a single interpretable measure while retaining the underlying domain-level information for further investigation.

The system also identifies the dominant source of relative vulnerability as:

- **Health-driven**
- **Education-driven**
- **Both equally**

This additional information helps users understand not only which districts receive higher priority, but also which domain requires greater attention.

D. Existing Gap in Decision-Making

The project was developed around several practical gaps in conventional district-level analysis.

Fragmented Data

Health and education information may exist in separate datasets. Analysing them independently makes it difficult to develop an integrated view of district conditions.

Different Indicator Scales

The indicators are measured using different units and ranges. Directly combining them without transformation would produce misleading results.

Different Indicator Directions



Some indicators represent better outcomes when their values are higher, while others represent greater vulnerability when their values are higher.

For example:

- Higher immunization → generally better
- Higher literacy → generally better
- Higher dropout → greater concern
- Higher anaemia → greater concern

A common vulnerability framework is therefore required.

Limited Prioritisation

Individual indicators can show problems, but they do not automatically provide an overall relative-priority framework.

Limited Action Orientation

A dashboard can successfully display information without necessarily helping users understand what the information means for prioritisation and resource allocation.

DHEIS attempts to address these gaps by connecting data preparation, analytical scoring, diagnosis and action-oriented reporting within one workflow.

E. Scope and Boundaries

The scope of DHEIS covers district-level analysis of selected health and education indicators. The health component uses selected indicators from **NFHS-5**, while the education component uses selected indicators from **DISE**. After data cleaning, reconciliation and integration, 602 districts were available for the core combined analysis.

The core prioritisation methodology uses **14 Tier 1 indicators**.

Health Indicators

The eight health indicators are:

1. Institutional Births (%)
2. Full Immunization (%)
3. Household Sanitation (%)
4. Health Insurance (%)
5. Stunted Children (%)
6. Wasted Children (%)
7. Underweight Children (%)
8. Child Anaemia (%)

Education Indicators

The six education indicators are:

1. Overall Literacy
2. Schools with Electricity (%)
3. Schools with Water (%)
4. Schools with Road Connectivity (%)
5. Dropout Rate (%)



6. True Pupil-Teacher Ratio

The project also contains additional indicators that are not included in the core Priority Index. These **Tier 2 variables** are retained for diagnostic analysis, contextual understanding and dashboard drill-downs.

The system is designed as a **decision-support tool**. It does not make autonomous government decisions, determine actual funding allocations or guarantee that a particular intervention will produce a specific outcome.

The priority classification represents **relative analytical priority within the project dataset**, rather than an absolute statement that one district is universally more developed or underdeveloped than another.

F. Data Quality as Part of the Project Journey

Data quality investigation was an important part of the DHEIS development process.

During the education-data analysis, the available pupil-teacher ratio field produced values that did not behave like a meaningful ratio. Instead of directly using the field, its underlying data was investigated.

The project identified that a reliable pupil-teacher ratio could be derived from the available enrollment and teacher information:

True Pupil-Teacher Ratio = Total Enrollment / Total Teachers

This was an important analytical correction because using an incorrectly interpreted field could have affected both the priority calculation and the machine-learning model.

The issue demonstrates an important principle followed throughout DHEIS:

Data should be investigated and validated before being used for decision-making.

This data-quality investigation is therefore considered an important part of the project's analytical journey rather than simply a preprocessing step.

G. Innovation Component

The main innovation of DHEIS is the integration of multiple analytical capabilities into one district intelligence workflow.

The project combines:

Health + Education + Data Engineering + Priority Analytics + Machine Learning + Business Intelligence + Generative AI

The DHEIS Priority Index provides a transparent scoring mechanism, while machine learning provides an additional validation layer. Power BI transforms the analytical outputs into

interactive dashboards, and Streamlit provides a more application-oriented interface for district exploration.

The generative-AI component further improves accessibility by converting structured analytical information into readable district briefs and district comparisons.



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However, the AI component is deliberately designed as an **interpretation assistant rather than an autonomous decision-maker**. It is constrained to the project data and instructed not to invent statistics, unsupported causes, policies or causal conclusions.

The overall innovation can therefore be represented as:

**Integrated Data → Validated Indicators → Priority Intelligence → ML Validation →
Interactive Analysis → AI-Assisted Interpretation → Action Support**

H. Overall Introduction to the DHEIS Solution

DHEIS was developed as an end-to-end system rather than as an isolated visualization or machine-learning exercise.

The complete solution follows the journey:

Raw Data



Clean and Validate



Integrate Health and Education Data



Transform Indicators into Vulnerability Scores



Calculate Health and Education Sub-Scores



Generate DHEIS Priority Index



Classify Priority Tiers



Identify Dominant Drivers



Validate Using Machine Learning



Present Results through Power BI



Explore Results through Streamlit



Generate AI-Assisted District Briefs



Support Resource Allocation and Intervention Planning

Through this approach, DHEIS demonstrates how data analytics can be transformed into a practical decision-support system for district-level planning.



3. PROJECT OBJECTIVES

A. Primary Objective

The primary objective of DHEIS (District Health & Education Intelligence System) is to develop an integrated district-level decision-support system that combines health and education indicators to identify relative priority, understand major areas of concern and support evidence-based resource allocation and intervention planning.

The project aims to transform raw district-level datasets into meaningful intelligence through a complete analytical workflow involving data engineering, data quality validation, priority scoring, machine-learning validation, interactive visualization and AI-assisted interpretation.

B. Data Engineering Objectives

The first objective is to establish a reliable and structured data foundation for the DHEIS system.

The project aims to:

- Collect district-level health and education data from relevant sources.
- Store and manage the raw datasets using PostgreSQL/Supabase.
- Process and transform the datasets using Databricks and Apache Spark/PySpark.
- Implement a Bronze-Silver-Gold data architecture.
- Preserve raw source information in the Bronze layer.
- Clean, transform and integrate the datasets in the Silver layer.
- Store business-ready analytical outputs in the Gold layer.
- Reconcile district-level records across the health and education datasets.
- Maintain data quality checks before using indicators for analysis.

The final integrated analytical dataset contains **602 districts** after the required reconciliation and integration process.

C. Data Quality and Validation Objectives

An important objective of the project is to ensure that the indicators used in the analysis are meaningful and correctly interpreted.

The project aims to:

- Identify missing, inconsistent and unexpected values.
- Examine indicator distributions before using them for modelling.
- Validate relationships between related variables.
- Investigate suspicious or incorrectly interpreted fields.
- Correct derived measures where necessary.



- Ensure that analytical indicators represent the intended concepts.

A key example was the investigation of the available pupil-teacher ratio field. Since the original values did not behave as a meaningful ratio, the project recalculated the measure using:

True Pupil-Teacher Ratio = Total Enrollment / Total Teachers

This ensured that the corrected indicator could be used more appropriately in the DHEIS analytical framework.

D. Analytical Objectives

The analytical objective is to convert multiple health and education indicators into an interpretable district-priority framework.

The project aims to:

- Select relevant health and education indicators.
- Separate core indicators from supporting contextual indicators.
- Normalize selected indicators onto a common **0–100 vulnerability scale**.
- Reverse higher-is-better indicators so that the direction of vulnerability remains consistent.
- Calculate a Health Sub-Score.
- Calculate an Education Sub-Score.
- Combine both domain scores into the DHEIS Priority Index.
- Classify districts into High, Medium and Low relative-priority tiers.
- Identify the dominant domain contributing to a district's relative vulnerability.

The core Priority Index uses **14 Tier 1 indicators**, consisting of eight health indicators and six education indicators.

The DHEIS Priority Index is calculated as:

DHEIS Priority Index = (Health Sub-Score + Education Sub-Score) / 2

The resulting framework provides a transparent method for comparing district-level relative priority.

E. Machine Learning Objectives

The machine-learning component is designed to provide an additional validation layer for the rule-based DHEIS priority classification.

The objectives are to:

- Use the original fourteen Tier 1 indicators as predictive features.
- Predict the DHEIS priority tier.
- Prevent data leakage by excluding derived scoring variables from the model features.
- Compare multiple classification algorithms.
- Use stratified train-test splitting because of the class distribution.
- Apply cross-validation and bounded hyperparameter tuning.
- Evaluate the models using classification performance metrics.



- Give particular importance to identifying High-priority districts correctly.
- Analyse model errors rather than relying only on overall accuracy.
- Track experiments and register the selected model using MLflow.

The models evaluated were:

- Dummy Classifier
- Logistic Regression
- Random Forest
- Gradient Boosting

The final selected model was a **tuned Logistic Regression model**, chosen with emphasis on reliable identification of the High-priority class.

F. Business Intelligence and Visualization Objectives

Another objective of DHEIS is to make analytical results accessible to users who may not work directly with raw datasets or machine-learning outputs.

The project aims to:

- Develop interactive Power BI dashboards.
- Display district priority distribution.
- Present health and education performance.
- Enable district and state-level comparisons.
- Highlight major indicators and drivers.
- Provide drill-down capabilities.
- Present action-oriented views.
- Support resource-allocation analysis.
- Provide urgency levels for potential intervention planning.
- Allow users to explore district conditions dynamically.

The Power BI solution therefore acts as the primary business-intelligence layer between the analytical data and decision-makers.

G. Application Development Objectives

The project also aims to provide an application-based interface through Streamlit.

The Streamlit application is designed to:

- Provide district-level overviews.
- Allow state and district filtering.
- Display the DHEIS Priority Index.
- Compare health and education performance.
- Benchmark a district against its state.
- Analyse equity and contextual indicators.
- Compare two districts.
- Identify the largest indicator gaps.
- Present AI-generated district briefs.
- Provide AI-assisted two-district comparisons.



This makes the project more interactive and exploratory than a static analytical report.

H. AI-Assisted Intelligence Objectives

The project incorporates generative AI to improve the interpretation of analytical results.

The objectives of the AI component are to:

- Convert structured district data into understandable analytical briefs.
- Summarize the overall condition of a selected district.
- Highlight key concerns.
- Identify priority areas.
- Generate data-grounded recommendations.
- Compare two districts using their available project data.
- Explain major differences between districts.

The AI component is implemented using the **Groq API** with the **openai/gpt-oss-20b** model.

The AI system is constrained to use only the information provided from the project. It is instructed not to invent statistics, unsupported causes, policies, targets or causal relationships.

Therefore, the AI feature is intended to provide **AI-assisted interpretation**, rather than autonomous policy-making.

I. Resource Allocation and Intervention Objectives

DHEIS aims to connect analytical findings with potential action.

The project therefore includes action-oriented analysis that helps users move from identifying a priority district to considering possible intervention areas.

The system provides concepts such as:

- Primary Concern
- District Intervention Plan
- Resource Allocation
- WHERE TO ALLOCATE
- WHAT TO FUND
- Urgency levels:
 - Immediate
 - High
 - Medium
 - Monitor
- Budget strategies:
 - Target High Need
 - Balance Needs
 - Equity First

These outputs are intended to support human decision-making rather than automatically determine actual government funding or intervention decisions.



J. Expected Outcomes and Deliverables

The expected outcomes of the project were to develop a complete and usable district intelligence solution rather than an isolated analysis.

The major deliverables include:

1. **Integrated District Dataset** containing the required health and education information for 602 districts.
2. **Bronze-Silver-Gold Data Pipeline** for structured data engineering.
3. **DHEIS Priority Index** for transparent district prioritisation.
4. **Health and Education Sub-Scores** for domain-level analysis.
5. **Priority Tier Classification** consisting of High, Medium and Low categories.
6. **Dominant Driver Analysis** identifying Health-driven, Education-driven or Both equally.
7. **Machine Learning Validation Model** for priority-tier prediction.
8. **MLflow Registered Model** named district_priority_classifier.
9. **Interactive Power BI Dashboard** for analytical reporting and decision support.
10. **Streamlit Application** for interactive district exploration.
11. **AI-Assisted District Briefs** for easier interpretation of analytical results.
12. **Two-District Comparison Capability** for comparative analysis.
13. **Action-Oriented Resource Allocation Views** for supporting intervention planning.
14. **GitHub Repository** containing the project implementation and supporting files.

K. Overall Objective-to-Outcome Relationship

The objectives of DHEIS can be summarised as a continuous transformation from data to decision support:

Objective Area	Intended Outcome
Data Collection	Reliable health and education datasets
Data Engineering	Structured Bronze-Silver-Gold pipeline
Data Quality	Validated and meaningful indicators
Priority Analytics	DHEIS Priority Index
Domain Analysis	Health and Education Sub-Scores
Classification	High / Medium / Low priority
Driver Analysis	Health / Education / Both equally
Machine Learning	Independent priority-tier validation
Business Intelligence	Interactive Power BI dashboard
Application	Streamlit district intelligence
Generative AI	Evidence-grounded analytical briefs
Action Support	Resource allocation and intervention planning

Overall, the project objectives were designed to ensure that DHEIS could move beyond data collection and visualization toward a complete analytical decision-support workflow.



4. METHODOLOGY & RESULTS

A. Methodology Overview

The methodology of DHEIS follows an end-to-end data analytics and decision-support workflow. The project was developed in multiple stages, beginning with raw health and education datasets and progressing through data validation, data engineering, analytical scoring, machine-learning validation, visualization, application development and AI-assisted interpretation.

The overall methodology can be represented as:

Data Collection → Data Storage → Data Quality Analysis → Data Cleaning → Data Integration → Feature Engineering → Priority Index → Machine Learning Validation → Business Intelligence → Application → AI Intelligence → Action Support

Each stage was developed to address a specific requirement of the final system.

The methodology was designed with two important principles:

1. **Transparency** — the core priority calculation should be understandable and reproducible.
2. **Actionability** — the final outputs should help users move from identifying a district to understanding its major concerns and considering appropriate resource allocation.

B. Data Collection and Source Integration

DHEIS combines information from two major domains: health and education.

Health Data

The health component uses district-level indicators from **NFHS-5 (National Family Health Survey – Round 5)**.

The health dataset contains indicators related to areas such as:

- maternal and child health,
- immunization,
- sanitation,
- health insurance,
- child nutrition,



- anaemia and other health-related conditions.

Education Data

The education component uses district-level information from **DISE (District Information System for Education)**.

The education data contains indicators related to:

- literacy,
- school infrastructure,
- electricity,
- water,
- road connectivity,
- dropout rates,
- enrollment,
- teachers and related education characteristics.

District Dimension

A district dimension was maintained to provide a structured reference for district-level integration and analysis.

The initial records across the main datasets were:

Dataset	Records
District Dimension	707
Raw Health	706
Raw Education	603

Final Integrated Dataset 602

The final integrated dataset was produced after district-level reconciliation and joining of the required health and education information.

One education record did not match during the final integration process and was therefore not included in the 602-district analytical dataset.



C. Data Storage and Engineering Architecture

The project uses **PostgreSQL/Supabase** for structured data storage and **Databricks** for data processing and transformation.

The data engineering architecture follows the **Bronze-Silver-Gold** approach.

Bronze Layer

The Bronze layer preserves the raw source data with minimal transformation.

Its purpose is to:

- maintain the original source information,
- provide traceability,
- preserve raw values,
- allow the data to be reprocessed when required.

The main raw health and education datasets were initially stored and examined at this stage.

Silver Layer

The Silver layer contains cleaned, validated and transformed data.

The major processing activities include:

- cleaning inconsistent values,
- handling data types,
- checking missing values,
- validating indicators,
- standardising district information,
- deriving corrected measures,
- integrating health and education datasets.

The integrated analytical dataset was developed in the Silver layer as:

silver_district_combined

Gold Layer

The Gold layer contains business-ready analytical outputs.

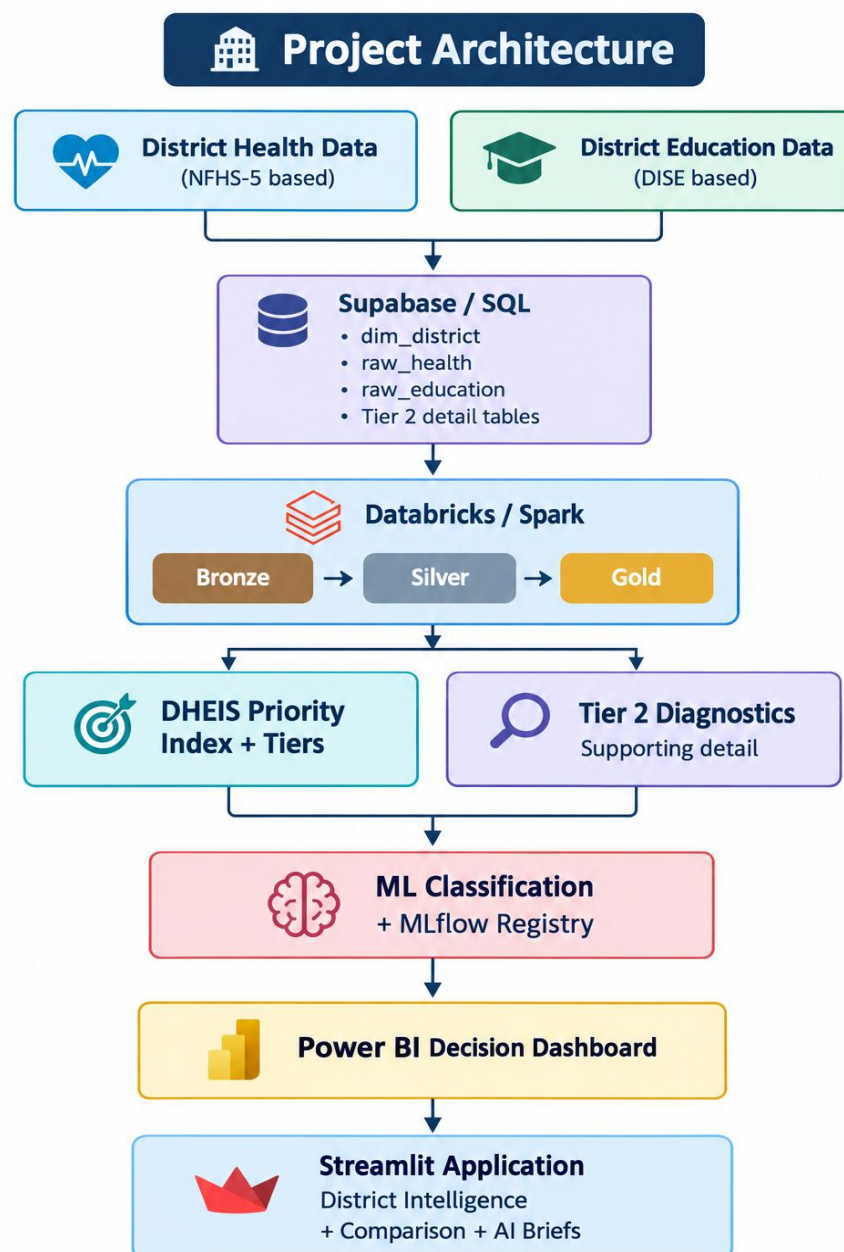
The main final analytical table is:



gold_district_priority

This layer contains the fields required for:

- Priority Index analysis,
- priority classification,
- driver analysis,
- machine-learning preparation,
- Power BI reporting,
- Streamlit application development.





D. Data Quality Investigation and Correction

Data validation was one of the most important stages of the project because the quality of the analytical output depends directly on the quality and interpretation of the input indicators.

During education-data analysis, an issue was identified with the available pupil-teacher ratio field.

The values in the original field did not behave like a meaningful pupil-teacher ratio. Instead of accepting the field directly, the relationship between enrollment and teacher information was investigated.

The project recalculated the measure using:

True Pupil-Teacher Ratio = Total Enrollment / Total Teachers

The corrected measure was then used as the education indicator in the DHEIS analytical framework.

This investigation was important because using an incorrectly interpreted indicator could have affected:

- the Education Sub-Score,
- the overall DHEIS Priority Index,
- district priority classification,
- machine-learning results,
- dashboard interpretation.

This stage demonstrated that data analytics is not only about applying formulas to datasets. It also requires understanding what each variable actually represents.

E. Selection of Analytical Indicators

The project separates indicators into **Tier 1** and **Tier 2** categories.

Tier 1 Indicators

Tier 1 contains the fourteen indicators used in the core DHEIS Priority Index and machine-learning model.

Health Indicators

The eight health indicators are:

1. Institutional Births (%)



2. Full Immunization (%)
3. Household Sanitation (%)
4. Health Insurance (%)
5. Stunted (%)
6. Wasted (%)
7. Underweight (%)
8. Child Anaemia (%)

Education Indicators

The six education indicators are:

1. Overall Literacy
2. Schools with Electricity (%)
3. Schools with Water (%)
4. Schools with Road Connectivity (%)
5. Dropout Rate (%)
6. True Pupil-Teacher Ratio

Tier 2 Indicators

Tier 2 indicators are not included directly in the core Priority Index.

Instead, they are used for:

- diagnostic analysis,
- contextual understanding,
- drill-down analysis,
- supporting dashboard interpretation,
- identifying additional characteristics of a district.

This separation prevents the core score from becoming unnecessarily complex while still retaining useful information for deeper analysis.

F. Indicator Direction and Vulnerability Transformation

The selected indicators do not all have the same interpretation.



Some indicators are **higher-is-better**, meaning that a higher value generally indicates better conditions.

Examples include:

- Institutional Births
- Full Immunization
- Household Sanitation
- Health Insurance
- Overall Literacy
- School Electricity
- School Water
- School Road Connectivity

Other indicators are **lower-is-better**, meaning that higher values represent greater vulnerability.

Examples include:

- Stunting
- Wasting
- Underweight
- Child Anaemia
- Dropout Rate
- True Pupil-Teacher Ratio

To make these indicators comparable, they were transformed into a common:

0–100 Vulnerability Scale

For higher-is-better indicators, the scale is reversed so that lower performance produces higher vulnerability.

For lower-is-better indicators, higher values naturally correspond to higher vulnerability.

This creates a consistent interpretation:

Higher vulnerability score = greater relative concern

The transformation allows indicators with different units and directions to contribute to a common priority framework.



G. DHEIS Priority Index Methodology

The central analytical component of the project is the **DHEIS Priority Index**.

The fourteen Tier 1 indicators are first converted to vulnerability scores.

The health indicators are then averaged to calculate:

Health Sub-Score

The six education indicators are averaged to calculate:

Education Sub-Score

The overall priority score is calculated as:

$$\text{DHEIS Priority Index} = (\text{Health Sub-Score} + \text{Education Sub-Score}) / 2$$

This equal-weighted approach ensures that health and education contribute equally to the overall priority score.

A higher Priority Index represents greater relative vulnerability across the combined health and education dimensions.

H. Priority Tier Classification

After calculating the DHEIS Priority Index for each district, districts are classified into three relative-priority categories.

Priority Tier Relative Position

High	Top 20%
Medium	Middle 60%
Low	Bottom 20%

The classification is based on the distribution of the calculated Priority Index values.

This means that the tiers represent **relative priority within the analysed dataset** rather than an absolute judgement about a district.

The classification allows users to quickly identify districts requiring greater analytical attention while still retaining the underlying numerical Priority Index.

I. Dominant Driver Analysis

DHEIS provides an additional interpretation layer by comparing the Health Sub-Score and Education Sub-Score.



The difference between the two domain scores is used to identify the dominant driver.

Health-Driven

A district is classified as **Health-driven** when health vulnerability exceeds education vulnerability by more than 5 points.

Education-Driven

A district is classified as **Education-driven** when education vulnerability exceeds health vulnerability by more than 5 points.

Both Equally

If the difference between the two scores is within 5 points, the district is classified as:

Both equally

This approach provides a simple explanation of where the relative vulnerability is concentrated.

J. Machine Learning Methodology

Machine learning was introduced as a validation layer for the rule-based DHEIS Priority Index.

The machine-learning task is a classification problem where the target variable is:

priority_tier

The model uses only the original fourteen Tier 1 indicators as input features.

The following derived variables were deliberately excluded:

- priority_index
- health_subscore
- education_subscore
- dominant_driver

These variables were excluded to prevent **data leakage**, because they are directly derived from the same scoring methodology used to create the target classification.

Models Evaluated

Four classification approaches were evaluated:

1. Dummy Classifier
2. Logistic Regression



3. Random Forest
4. Gradient Boosting

The dataset was divided using a **stratified 80/20 train-test split** so that the priority classes remained represented in both datasets.

A **5-fold stratified cross-validation** approach was used during model development and tuning.

Bounded hyperparameter tuning was applied to improve model performance while keeping the modelling process controlled.

The final test set remained untouched during model development and was used only for final evaluation.

K. Machine Learning Results

The final selected model was a **tuned Logistic Regression model**.

The model was selected with particular attention to the identification of High-priority districts.

The final held-out test results were:

Metric	Result
Test Accuracy	95%
High-tier Precision	96%
High-tier Recall	96%
High-tier F1-score	96%
Test Districts	121

The results indicate strong predictive performance on the held-out test dataset.

The High-tier recall of 96% is particularly relevant because failing to identify a genuinely High-priority district could be more important from a decision-support perspective than a small number of classification errors in lower-priority categories.

L. Baseline and Production Comparison

A majority-class Dummy Classifier was used as a baseline for comparison.

The baseline achieved:



- **59.5% accuracy**
- **0% High-tier recall**

This demonstrates that the final model provides substantially more useful classification performance than simply predicting the most common class.

The production comparison was also performed between the calculated DHEIS priority tier and the machine-learning prediction.

Out of the 602 integrated districts:

- **577 districts matched**
- **25 districts differed**

This comparison provides an additional way to examine where the machine-learning model agrees with or differs from the transparent analytical scoring system.

The differences were treated as useful points for investigation rather than automatically assuming that the machine-learning prediction should replace the rule-based result.

M. MLflow Model Tracking

MLflow was used to track the machine-learning experiments and manage the selected model.

The final registered model is:

district_priority_classifier

MLflow provides a structured way to maintain information related to the model-development process and supports reproducibility and model management.

The use of MLflow also strengthens the project's MLOps component by connecting experimentation with a registered production-oriented model.

N. Power BI Intelligence Layer

Power BI was used to convert the analytical outputs into an interactive business-intelligence dashboard.

The dashboard was designed to support both high-level monitoring and detailed district investigation.

Major dashboard capabilities include:

- Key Performance Indicators



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- Priority distribution
- Health performance
- Education performance
- District comparisons
- State comparisons
- Indicator analysis
- Driver analysis
- Drill-down analysis
- Action Queue
- Primary Concern
- District Intervention Plan
- Resource Allocation
- “WHERE TO ALLOCATE”
- “WHAT TO FUND”

The dashboard also includes urgency categories:

- Immediate
- High
- Medium
- Monitor

and budget strategies such as:

- Target High Need
- Balance Needs
- Equity First

These features transform the dashboard from a simple reporting interface into a decision-support layer.

O. Streamlit Application

A Streamlit application was developed to provide an interactive district intelligence interface.



The application allows users to explore district-level information using filters and analytical views.

Major features include:

District Overview

Users can select a district and view its overall Priority Index, priority tier and major analytical information.

Health and Education Analysis

The application allows users to examine the relative performance of health and education dimensions.

District-to-State Benchmarking

A district can be compared with its state-level context to identify areas where the district differs from the broader state picture.

Equity and Context Analysis

Additional indicators can be examined to provide context beyond the core Priority Index.

Two-District Comparison

Users can select two districts and compare their:

- overall priority,
- health and education conditions,
- indicator gaps,
- major concerns,
- priority areas.

Largest Indicator Gaps

The application highlights the indicators with the largest differences between the selected district and its comparison context.

These features make Streamlit a complementary application layer to the Power BI dashboard.

P. AI-Powered Intelligence

A generative-AI component was integrated into the Streamlit application using the **Groq API** and the **openai/gpt-oss-20b** model.



The purpose of the AI component is to convert structured analytical information into understandable written intelligence.

District AI Brief

For an individual district, the AI produces four sections:

1. Overall Assessment
2. Key Concerns
3. Priority Areas
4. Recommendations

Two-District AI Comparison

For two selected districts, the AI produces:

1. Overall Assessment
2. Key Differences
3. Priority Areas
4. Recommendations

The prompts used for the AI component contain important safeguards.

The AI is instructed to:

- use only the supplied project data,
- avoid inventing statistics,
- avoid unsupported causes,
- avoid unsupported government policies,
- avoid unsupported targets,
- avoid causal claims,
- provide district-level recommendations,
- use decision-support language.

Therefore, the AI component is intended to assist users in interpreting existing analytical evidence rather than independently determining policy or resource decisions.



Q. Resource Allocation and Intervention Planning

A major feature of the DHEIS methodology is the transition from analysis to potential action.

The system does not stop after identifying a High-priority district. Instead, the analytical outputs are used to identify areas where attention may be required.

The workflow is:

Priority Identification → Driver Identification → Indicator Diagnosis → Primary Concern → Intervention Planning → Resource Allocation

The Power BI action-oriented views provide concepts such as:

WHERE TO ALLOCATE

This focuses attention on districts or areas requiring greater relative priority.

WHAT TO FUND

This connects identified concerns with potential areas of resource focus.

District Intervention Plan

This provides a structured view of the main areas that may require attention.

The budget strategies provide different approaches depending on the decision context:

- **Target High Need** — concentrate resources on the highest-priority areas.
- **Balance Needs** — distribute resources across multiple areas of need.
- **Equity First** — give greater consideration to areas facing comparatively greater vulnerability.

These are analytical decision-support strategies and should not be interpreted as automatically generated government funding decisions.

R. Final Project Working Screenshots and Results

The final report should include screenshots of the completed DHEIS system to demonstrate the actual working output of the project.



Figure 4.1 — DHEIS Overall Power BI Dashboard

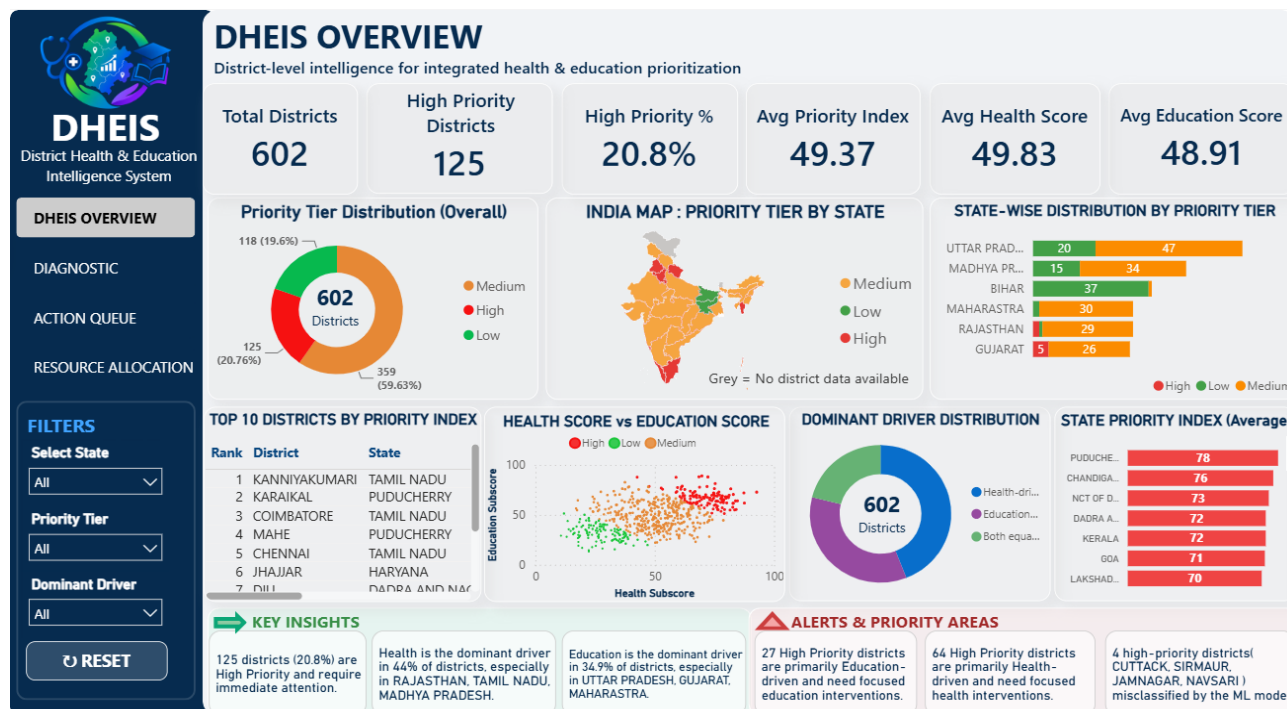


Figure 4.1: Overall DHEIS dashboard showing district priority distribution and key health and education indicators.

Figure 4.2 — District Priority and Analysis View

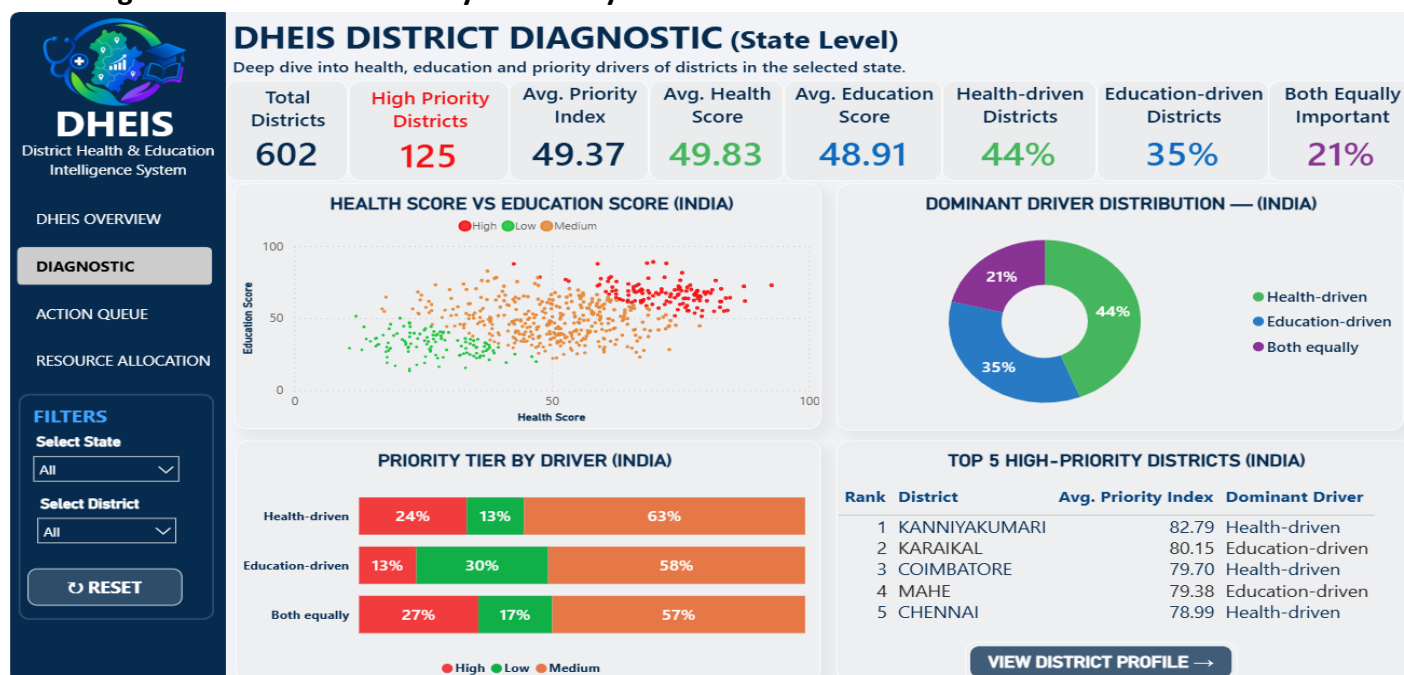


Figure 4.2: District-level priority analysis showing the DHEIS Priority Index, priority tier and supporting indicators.



Figure 4.3 — Action Queue / Intervention Planning

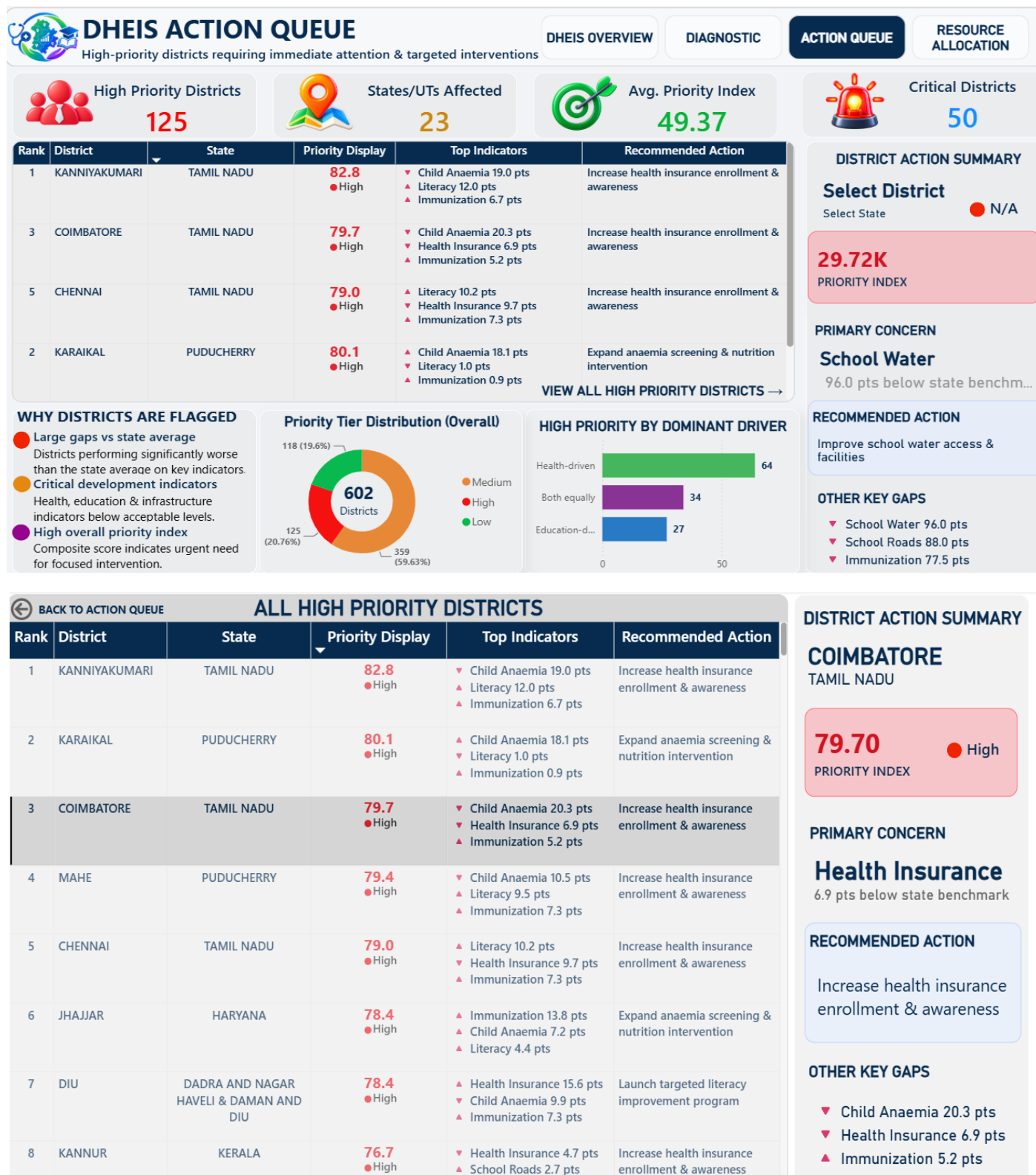


Figure 4.3: Action-oriented DHEIS view showing priority concerns, urgency and intervention planning.



Figure 4.4 — Resource Allocation View

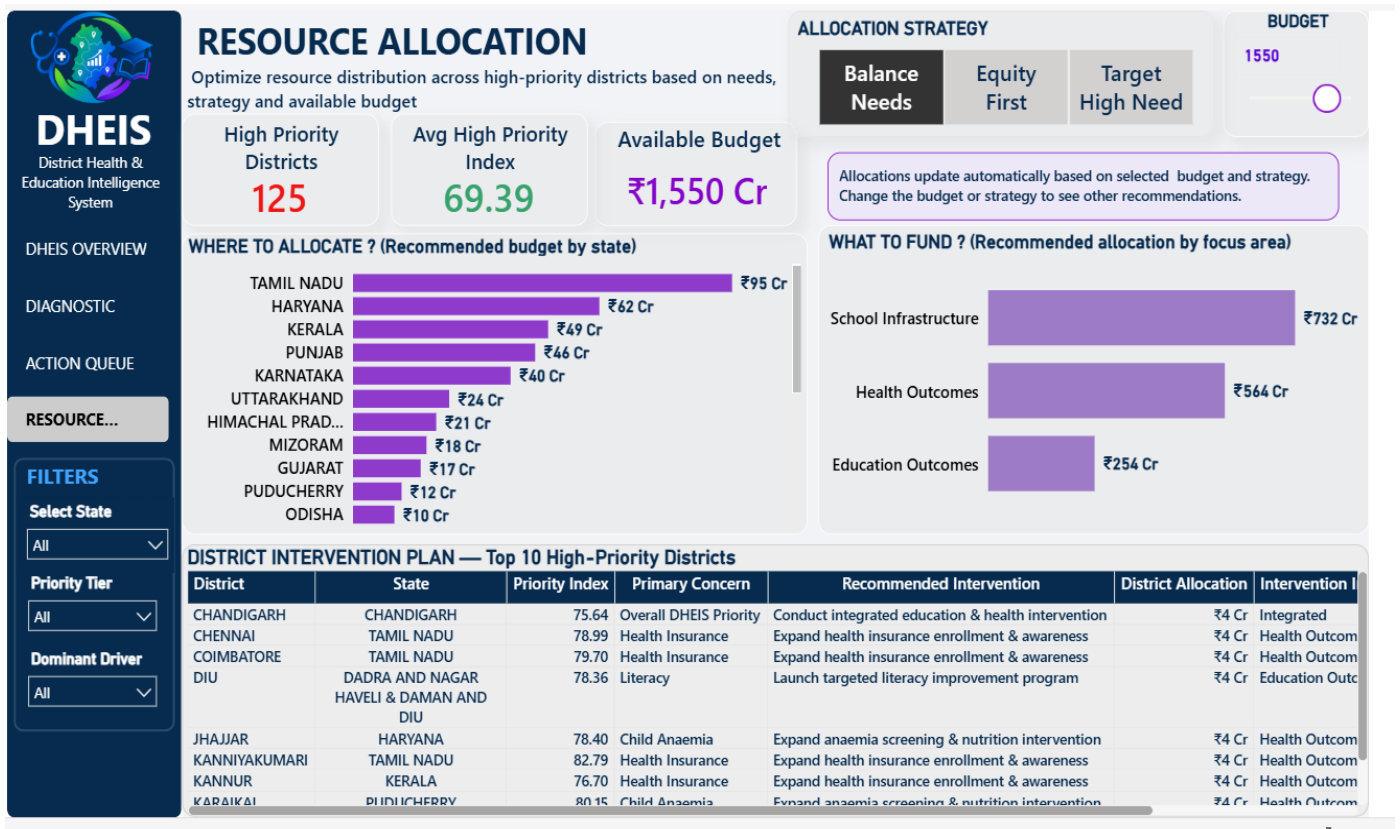


Figure 4.4: Resource allocation analysis showing “WHERE TO ALLOCATE” and “WHAT TO FUND” decision-support views.

S. Overall Methodology Outcome

The methodology resulted in an integrated district intelligence system that connects data engineering, analytics, machine learning and decision-support interfaces.

The completed workflow can be summarised as:

NFHS-5 + DISE



PostgreSQL / Supabase



Databricks + PySpark



Bronze Layer





Silver Layer



Gold Layer



14 Tier 1 Indicators



0–100 Vulnerability Transformation



Health + Education Sub-Scores



DHEIS Priority Index



High / Medium / Low Priority



Dominant Driver



Machine Learning Validation



MLflow Model



Power BI Dashboard



Streamlit Application



Groq AI Intelligence



Action Queue + Resource Allocation

This methodology demonstrates the complete transformation of raw public-sector data into an integrated analytical decision-support system.



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The results show that DHEIS is capable of combining multiple data sources, identifying relative district priority, validating the classification using machine learning, presenting the results interactively and generating evidence-grounded analytical explanations.

The project therefore achieves its central methodological goal:

to transform district-level health and education data into structured intelligence that can support prioritisation, diagnosis and resource-allocation thinking.

T. GitHub Repository

The complete project implementation, supporting files and application code are maintained in the project's GitHub repository.

The repository contains the project components required to demonstrate the development journey, including relevant data-processing, analytical, machine-learning and application files.

GitHub Repository: <https://github.com/sure-trust/MANPREET-KAUR-g1-data-analytics>

Streamlit Application: [DHEIS APP](#) and [simple DHEIS](#)



5. SOCIAL/INDUSTRIAL RELEVANCE OF PROJECT

A. Introduction

Data has become an important resource for decision-making in many sectors. However, data alone cannot create value unless it is transformed into meaningful information that supports action. In sectors such as health and education, large amounts of information are collected regularly, but decision-makers may still face difficulties in identifying priority areas when indicators are spread across multiple datasets.

DHEIS (District Health & Education Intelligence System) was developed to address this challenge by integrating health and education information into a single analytical framework. The system transforms raw district-level data into understandable intelligence through data engineering, analytical scoring, machine learning, interactive dashboards and AI-assisted interpretation.

The practical relevance of DHEIS lies in its ability to simplify complex information and support evidence-based thinking.

B. Social Relevance

Health and education are two of the most important pillars of human development. Improvements in these areas can directly influence quality of life, opportunities and long-term social progress.

Many districts may perform well in some indicators while facing challenges in others. Without a structured analytical framework, it can be difficult to identify where greater attention may be needed.

DHEIS contributes socially by:

- Combining health and education information into one view.
- Identifying districts with relatively greater vulnerability.
- Highlighting major concerns affecting districts.
- Supporting more informed discussions about resource allocation.
- Providing understandable information through dashboards and AI-generated briefs.
- Making analytical information easier to access for different types of users.

The project therefore demonstrates how data analytics can support public-interest decision-making.

C. Relevance to Health Sector

The health component of DHEIS includes indicators such as:

- Institutional Births
- Full Immunization



- Household Sanitation
- Health Insurance
- Child Stunting
- Child Wasting
- Underweight Children
- Child Anaemia

By combining these indicators into a Health Sub-Score, the system helps users identify districts where health-related concerns are comparatively higher.

Potential uses include:

- Identifying districts requiring greater analytical attention.
- Monitoring district-level health conditions.
- Supporting discussions related to health planning.
- Comparing district performance.
- Understanding major health vulnerabilities.

The system does not replace expert judgement or government planning processes, but it can support them with structured evidence.

D. Relevance to Education Sector

Education indicators included in DHEIS are:

- Overall Literacy
- School Electricity
- School Water Availability
- Road Connectivity
- Dropout Rate
- True Pupil-Teacher Ratio

These indicators are combined to create the Education Sub-Score.

Potential applications include:

- Identifying districts with education-related concerns.
- Comparing education infrastructure conditions.
- Understanding dropout-related challenges.
- Supporting district-level education analysis.
- Identifying areas where educational resources may require attention.

The project demonstrates how educational data can be transformed into decision-support information.

E. Importance for Data Analytics Industry

DHEIS is not only a social-impact project but also an example of a modern data analytics workflow.

The project combines:



- Data Collection
- Data Cleaning
- Data Validation
- SQL
- PostgreSQL / Supabase
- Databricks
- Apache Spark / PySpark
- Delta Lake
- Feature Engineering
- Machine Learning
- MLflow
- Power BI
- Streamlit
- Generative AI
- GitHub

This reflects the complete lifecycle of a real-world analytics project.

The project demonstrates that data analytics is not limited to visualization. It includes:

Data Engineering → Data Science → Business Intelligence → Application Development → AI Integration

This makes DHEIS relevant to industry practices where multiple technologies are combined into one solution.

F. Relevance to Public Administration and Planning

Decision-makers often need to determine:

- Which districts require greater attention?
- What are the major concerns?
- Where should resources be focused?
- Which areas require immediate intervention?

DHEIS attempts to support these questions through:

- Priority classification
- Driver analysis
- Action Queue
- District Intervention Plans
- Resource Allocation views
- Budget strategy suggestions

The system provides analytical support rather than making automatic decisions.

Human experts remain responsible for final planning and implementation.



G. Importance of Machine Learning Validation

The machine-learning component demonstrates how predictive analytics can be used alongside transparent rule-based methods.

The ML model:

- Uses only original indicators.
- Avoids data leakage.
- Validates the priority classification.
- Provides additional confidence in the analytical framework.

The selected Logistic Regression model achieved:

- 95% Accuracy
- 96% Precision for High-priority districts
- 96% Recall for High-priority districts
- 96% F1-score for High-priority districts

These results show that the identified patterns are sufficiently consistent for predictive modelling.

H. Importance of Generative AI Integration

The project also demonstrates a practical use of generative AI.

The AI component converts structured district data into:

- District briefs
- Two-district comparisons
- Key concern summaries
- Recommendation-oriented explanations

The AI is constrained to:

- Use project data only.
- Avoid unsupported claims.
- Avoid invented statistics.
- Avoid causal assumptions.

This demonstrates responsible use of AI in analytical applications.

I. Learning Outcomes from the Project

The project provided experience in:

Technical Learning

- SQL
- PostgreSQL



- Databricks
- PySpark
- Delta Lake
- Python
- Machine Learning
- MLflow
- Power BI
- Streamlit
- Generative AI Integration
- GitHub

Analytical Learning

- Data quality analysis
- Feature engineering
- Indicator selection
- Priority scoring
- Dashboard design
- Model evaluation
- Business problem solving

Professional Learning

- End-to-end project execution
 - Documentation
 - Problem solving
 - Data validation
 - Presentation skills
 - Version control
 - Decision-support thinking
-

J. Future Scope

The DHEIS framework can be expanded in several ways.

Possible future enhancements include:

1. Addition of more recent datasets.
2. Inclusion of more district-level indicators.
3. Real-time data integration.
4. Advanced geospatial analysis.
5. Predictive trend analysis.
6. Scenario simulation.
7. Mobile application development.
8. More advanced AI capabilities.
9. Policy impact assessment modules.
10. Cloud-based deployment.



The framework can also be adapted to other domains where multiple indicators need to be combined into a priority-based decision-support system.

K. Overall Relevance

DHEIS demonstrates how modern data technologies can be used to transform complex information into understandable intelligence.

The project combines:

Data Engineering + Analytics + Machine Learning + Visualization + AI

to create an integrated district intelligence system.

Its relevance lies not only in the technology used but also in the ability to support:

- Prioritisation
- Diagnosis
- Comparison
- Resource-allocation thinking
- Evidence-based discussions

The project therefore represents a practical example of how data analytics can be applied to real-world challenges in health and education.



6. LEARNING AND REFLECTION

A. Technical Learning

Working on DHEIS helped me gain practical experience in different areas of data analytics and data engineering. I worked with **SQL, Python, PostgreSQL, Databricks, PySpark, Delta Lake, Power BI, and Streamlit** during different stages of the project.

I learned how to work with raw datasets, perform data cleaning and validation, combine health and education data, and prepare the data for analysis. I also gained a better understanding of the **Bronze, Silver, and Gold data architecture** and how each layer is used in a data pipeline.

B. Data Analysis and Problem-Solving Learning

A major learning from the project was that data needs to be checked carefully before using it for analysis. During the project, I identified an issue with the original pupil-teacher field because its values did not behave like a normal ratio.

I investigated the issue and calculated the **True Pupil-Teacher Ratio** using total enrollment and total teachers. This helped me understand the importance of checking data meaning, not just its format or values.

I also learned how to select relevant indicators, convert them into a common vulnerability scale, and combine them to create the **DHEIS Priority Index**.

C. Machine Learning and MLOps Learning

The machine learning part of the project helped me understand the practical workflow of building a classification model. I learned about **train-test splitting, stratified cross-validation, model comparison, hyperparameter tuning, and evaluation metrics**.

I worked with different classification models and compared their performance before selecting the final model. I also gained practical experience with **MLflow** for tracking experiments and registering the selected model.

An important learning was understanding **data leakage** and ensuring that calculated priority outputs were not directly used as model features.

D. Power BI and Application Development

Through the Power BI part of the project, I improved my understanding of **DAX, interactive filters, KPIs, charts, comparisons, drill-downs, and dashboard design**. I learned how to present analytical results in a way that is easier for users to understand.



I also developed a **Streamlit application** that provides district-level analysis, comparisons, indicator gaps, and AI-generated briefs. This helped me understand how analytical results can be converted into a simple user-facing application.

E. Generative AI Learning

The integration of the **Groq API** introduced me to the practical use of generative AI in a data analytics project. I learned how to design prompts that provide the model with relevant project data and instructions for generating district briefs and comparisons.

I also learned why AI outputs need to be controlled so that the model does not create unsupported statistics, causes, or recommendations. In DHEIS, AI is therefore used to assist with interpretation rather than replace human decision-making.

F. Professional Learning

The project improved my ability to work through problems step by step and manage different technologies within one project. I also learned the importance of documenting changes, checking results, troubleshooting errors, and explaining technical concepts in a simple way.

Working on the complete project also helped me understand how different parts of a data analytics solution—**data engineering, analysis, machine learning, visualization, and application development**—work together.

G. Overall Experience and Reflection

Developing DHEIS provided experience in combining multiple technologies into one complete solution.

The project demonstrated that successful data analytics requires more than technical implementation. It requires:

Data Understanding + Data Quality + Analytical Thinking + Problem Solving + Communication

The project also showed the importance of validating data before analysis, designing metrics around a clear objective and presenting results in a way that users can understand.

Overall, DHEIS provided practical experience in transforming raw data into structured intelligence and strengthened the ability to develop an end-to-end data-driven solution.



7. FUTURE SCOPE & CONCLUSION

A. Conclusion

DHEIS (District Health & Education Intelligence System) was developed to combine district-level health and education data and convert it into useful information for identifying priority areas.

The project covered the complete data analytics process, including data collection, cleaning, validation, transformation, analysis, machine learning, dashboard development, and application development. The DHEIS Priority Index was created using selected health and education indicators to classify districts into High, Medium, and Low priority categories.

The project also included a machine learning model for priority classification, Power BI dashboards for analysis and visualization, and a Streamlit application for district-level exploration. A Groq-based AI feature was added to generate short, data-based district briefs and comparisons.

During the project, data quality issues were also identified and addressed. One important example was the incorrect interpretation of the original pupil-teacher field, which was investigated and replaced with a calculated pupil-teacher ratio using total enrollment and total teachers.

Overall, the project helped demonstrate how different data analytics and technology tools can be combined to build a practical district-level decision-support system.

B. Final Project Value

The main outcome of DHEIS is a single system that helps users:

- View health and education conditions at the district level.
- Identify districts requiring greater attention.
- Understand the main health or education drivers.
- Compare districts and states.
- View supporting indicators through dashboards and the Streamlit application.
- Use AI-generated summaries for easier interpretation.
- Support resource allocation and intervention planning based on the available data.

The system is intended as a **decision-support tool** and does not replace human decision-making.



C. Future Scope

The project can be further improved in the following areas:

- **More Recent Data:** Integrating newer health and education datasets as they become available.
- **Time-Series Analysis:** Adding multiple years of data to study changes in district performance over time.
- **More Indicators:** Including additional indicators when reliable and relevant data is available.
- **Improved ML Models:** Testing additional models and monitoring their performance with new data.
- **Automated Data Pipeline:** Automating data collection and updating of the Databricks layers.
- **Advanced AI Features:** Expanding AI assistance for district comparison, trend explanation, and report generation while keeping the outputs grounded in the available data.
- **Wider Deployment:** Making the system accessible to a larger number of users through a properly deployed web application.

D. Overall Conclusion

DHEIS successfully combines **data engineering, data analytics, machine learning, business intelligence, and generative AI** into one project. The main learning from the project was that reliable decision support depends not only on visualization, but also on proper data validation, meaningful indicator selection, and careful interpretation of results.

The project provides a practical foundation that can be extended further with better data, automation, and additional analytical capabilities.